Introduction to Data Science (CS 418)

UIC College of Engineering, 3 undergraduate hours, 4 graduate hours

Fall 2026

I. Instructor & Course Details

Jack Bandy

Email address: jxb@uic.edu

Drop-In Office Hours (in-person or virtual): Tuesdays 1:30 - 2:30 PM, other times via scheduling link in Canvas

Drop-In Hours location: CDRLC 3454

Teaching Assistants

Maksym Turkot (Maks)

Email address: mturk5@uic.edu

Drop-In Office Hours (in-person or virtual): Tuesdays and Wednesdays, 12:30 - 1:30 PM

Drop-In Hours location: CDRLC 2404 (The 404)

Course Site: dodatascience.fun

The course site includes slides, the syllabus, worksheets, the course schedule, and links to various reference materials.

For technical questions about Canvas, email the Learning Technology Solutions team at LTS@uic.edu.

Email Expectations

Students are responsible for all messages sent to UIC email and Canvas accounts. I generally aim to respond to emails within 24 hours during weekdays. I generally do not check my email on Saturday and Sunday.

PrairieLearn: us.prairielearn.com/pl/course_instance/216433 — homework questions, quizzes, and the final exam.

Canvas: canvas.uic.edu — grades, coding exercises, and project scaffolding. Please log in regularly to monitor your grades and to learn about any developments related to the course.

Course Modality and Schedule

Meeting time:

- Monday and Wednesday
- 3:30 - 4:45 PM

Location:

- Lecture Center Building C
- Room C001

CRN: 44033

Schedule Type: Lecture

Instructional Method: Meet on campus (in-person)

II. Course Information

Catalog Course Description

Provides an in-depth overview of data science in engineering. Topics include modeling, storage, manipulation, integration, classification, analysis, visualization, information extraction, and big data in the engineering domain.

Prerequisite(s): Grade of C or better in CS 251; and STAT 381 or IE 342 or ECE 341.

Section-Specific Description

Although it may or may not be the "sexiest job of the 21st century," data science remains critically important for knowledge work, evidence-based decision-making, planning, and collaboration across society. The purpose of this course is to become acquainted with the "data science lifecycle" and practice navigating it effectively. The data science lifecycle includes: asking good questions, obtaining and stewarding data, making sense of data, understanding the world based on data, and communicating insights from data. "Data science" is a broad and ever-expanding family of topics, many of which can be technically complex and challenging. We will work together in this class to develop the skills and knowledge needed for effective data science work, including a "growth mindset" – an attitude that will allow you to continue learning.

Welcome!

Data are fun, and data science can be even more fun! Although I have taught basically every individual topic in this course at one point or another, this is my first time teaching them all under the umbrella of a full-semester data science course. This means there will probably be a few hiccups and wrinkles to iron out this semester - thank you in advance for your patience and your feedback to help me improve the class for future semesters!

Growth Mindset:

Course materials and exercises might be complex and challenging, but they are crucial to your intellectual and personal growth and development. Just like a challenging weight-lifting exercise is intended to strengthen your body, a challenging class exercise is intended to strengthen your mind.

As will be discussed in class, I try to keep exercises in the "zone of proximal development." This is a concept from educational psychology which basically says an exercise should not be too easy or too difficult. Again, this is kind of analogous to weight-lifting exercises: there is a "goldilocks" zone that is challenging enough to help you learn and grow, but not so challenging as to discourage you.

If and when you need help, please reach out sooner rather than later! Your feedback is the best way for me to know if we are in the "zone of proximal development," or if something is too easy or too difficult.

Learning Priorities (i.e. Course Goals)

If we are successful, by the end of the semester, you will have proficiency in the following: - Obtaining and preparing data for exploratory data analysis, including intentional selection of data storage formats and "data minimization" techniques - Techniques for exploratory data analysis (EDA), including summary statistics, tabulation, aggregation, and descriptive visualizations - Creating "persuasive" data visualizations to communicate specific patterns, trends, and/or insights from data - Developing models for data analysis (i.e. regression, classification, clustering) - Evaluating a model's performance and understanding its structure, meaning, statistical significance, and limitations - Communicating the results and implications of data analyses, and explaining how those results are connected to various decisions made throughout the data science lifecycle

Course Topics

- Course introduction, the data science lifecycle, asking questions
- Statistics review, Python foundations, dataframes, Polars, and obtaining data
- Data wrangling, filtering, and data formats
- Exploratory analysis and descriptive statistics
- Visualization basics, chart choice, visual encodings, design, communication, and critique
- Hypothesis testing, estimation, sampling, and randomness
- Linear regression and classification
- Decision trees, support vector machines, and kernels
- Clustering, principal component analysis, and model evaluation
- Recommendation systems and A/B testing
- Network analysis, graphs, relationships, and network measures

Course Texts and Materials

For Fall 2026, supporting texts and materials will be open-source and freely available.

Required Technology: You will need access to a computer capable of running Jupyter notebook and Python, as well as PrairieLearn and Canvas.

All required software is free. If that ever seems to not be the case, please let me know as soon as possible.

Also, if access to a suitable computer or internet connection presents a challenge for any reason, please tell me as soon as possible — this is a solvable problem and you will not be the first person to ask.

Core Texts: There is no required textbook to purchase. The course draws on three freely available open-access texts:

- *Learning Data Science* by Sam Lau, Joey Gonzalez, and Deb Nolan
- *Data Science: A First Introduction with Python* by Tiffany Timbers, Trevor Campbell, Melissa Lee, Joel Ostblom, and Lindsey Heagy
- *The Art of Data Science* by Roger D. Peng and Elizabeth Matsui

Readings from these texts will be listed alongside the weekly schedule. If obtaining any course material presents a challenge for any reason, please tell me as soon as possible.

Respect for Copyright

Slides and other reference materials are publicly available online under a Creative Commons license.

Please protect the integrity of other course materials and content: as a general rule of thumb, please do not upload course materials you did not create onto third-party websites, and do not share private content with anyone who is not enrolled in our course. This includes worksheets, rubrics, and any readings posted behind the Canvas login. If you have any questions about this, please ask.

The work you create for this course belongs to you. I may ask you to submit some exercises publicly through the course website, and I may ask to reuse your work as an example for future classes. You can always say "no" and choose to submit your work privately (i.e. only through Canvas).

Content Notices

This is a 400-level University course; as such, we will aim to discuss all topics with maturity, dignity, humanity, rigor, and respect.

Data science is interdisciplinary and touches many different topics; some materials will address sickness, death, war, politics, discrimination, surveillance, sex, violence, and related subjects which may be distressing. Please know that the goal is never to cause distress, but to meaningfully engage with real-world issues.

Furthermore, some readings and other materials may expose you to ideas, subjects, or views that challenge you, cause you discomfort, or recall past negative experiences or traumas. I try to include content notices whenever appropriate. If you have suggested changes (i.e. if there is a place where you recommend adding a content notice), or you would like me to be aware of a specific topic of concern, please email or visit me during office hours.

III. Course Policies & Classroom Expectations

Grading Policy and Point Breakdown

Assessments

In order to provide a final letter grade at the end of the class, I will use the following combination of graded exercises:

Category	Weight
In-Class Worksheets	**20%**
One per class	
In-Class Quizzes	**20%**
Every other week, seven total	
Group Project	**20%**
Question memo	
Data acquisition	
EDA	
Data documentation	
Rhetorical analysis	
Public repository	
Written report	
Presentation	
Peer assessment	
PrairieLearn Homeworks	**15%**
One per week, unlimited retakes	
Coding Exercises	**15%**
Final Exam	**9%**
Collective Attendance	**1%**

Further details and rubrics for each exercise will be published on Canvas.

Grading Scale

You can expect the following correspondence between your percentage grade in the course and your final letter grade, and the "equivalent" description from the registrar's grading system.

F	D	C	B	A
59% or below Failure	60–69% Poor but passing	70–79% Average	80–89% Good	90–100% Excellent

Student Accountability for Coursework

Instructions for every graded exercise will be posted before the work is due, usually at least one week ahead. Some exercises are still being developed, so please let me know as soon as possible if you notice any issues.

Unless an exercise says otherwise:

- **In-class worksheets** are completed in class, and submitted in canvas via picture.
- **In-class quizzes** are taken during class on your laptop.
- **PrairieLearn homeworks** are submitted in PrairieLearn by the posted due date. There are unlimited retakes.
- **Coding exercises** and **group project milestones** are submitted in Canvas by the posted due date.

Due dates for everything submitted online are visible in PrairieLearn and Canvas, and I plan to highlight them in class. See "Policy for Missed or Late Work" below for what happens when something is submitted late.

Midterm Grades

Grades in Canvas are kept current throughout the semester, so you can check where you stand at any point.

Around the midpoint of the semester I will also do a check-in: if your grade suggests you are at risk of not passing, I will email you directly so we have time to intervene. You are always welcome to come to drop-in hours to ask about your grade.

Final Exams

The final exam will be administered in PrairieLearn and is worth 9% of your grade. Most of the final exam questions will be "variants" of the same questions you will have already seen in homework and quizzes throughout the semester. This means that steady work on the homework and quizzes is excellent preparation for the final.

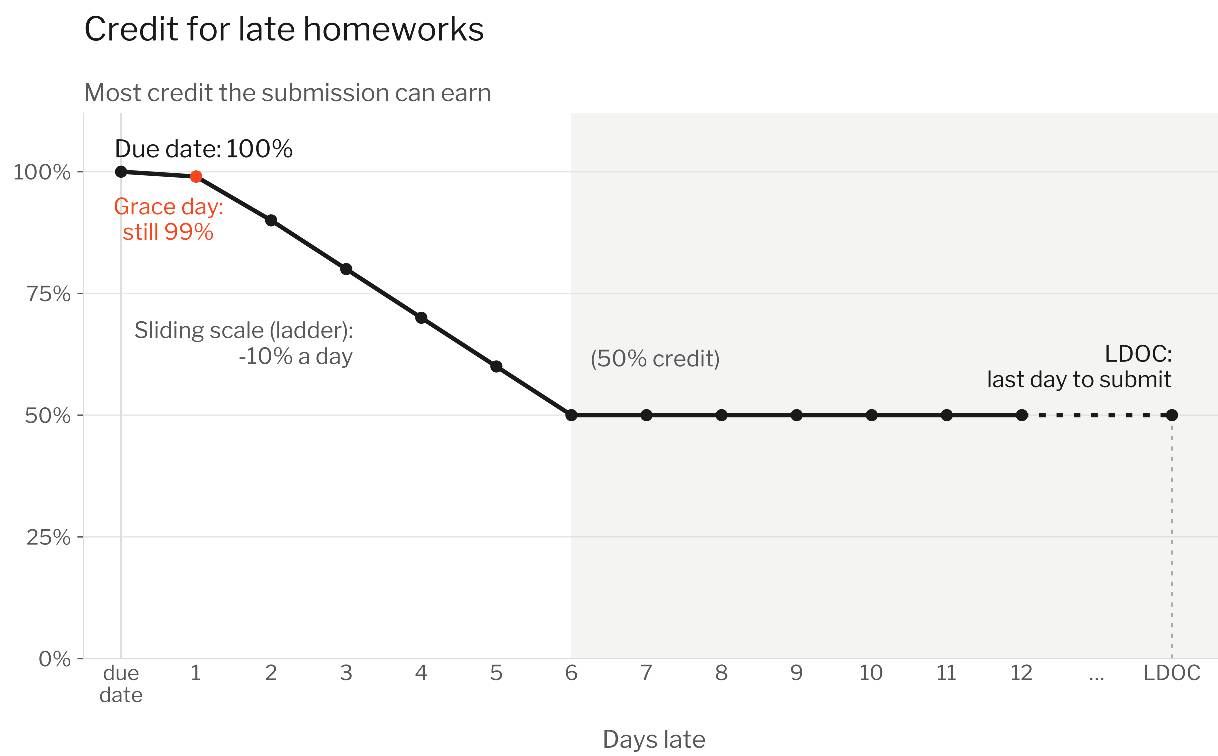
Scheduling follows the registrar's schedule of final exams and exam policies.

Policy for Missed or Late Work

Due dates will be visible in PrairieLearn (Homework Questions) and/or Canvas (Coding Exercises, Project Scaffolding), and highlighted in class. Work submitted up to one day late still earns effectively full credit (99%). After that, an automatic "sliding scale" deducts 10% for each additional day late, down to a floor of 50% credit.

Anything submitted more than one week after the original due date will only receive 50% credit, and nothing will be accepted after the last day of class.

If you anticipate any difficulty meeting a due date, contact the instructor as soon as possible. I am often able to adjust due dates, especially if multiple people want/need an extension.



Credit for late homeworks: 99% on the one-day grace period, then a 10%-a-day sliding scale down to a 50% floor, until the last day of class.

Adapted from the UIC CATE syllabus template.

Attendance / Participation Policy

The activities we do during class are essential to your learning in this course, so please make every effort to attend all class meetings and to arrive to class on time.

I recognize that illness, personal emergencies, university obligations, religious observances, and other circumstances may sometimes cause you to be late to class or prevent your attendance entirely.

An audacious experiment. Attendance is worth 1% of your grade, and it is a collective 1%. Each class meeting, after the agenda slide, I will draw twelve names at random using live random numbers from random.org. Everyone in the class receives the same attendance grade out of ten for that day, based on how many of the drawn students are present.

There are two motivations for this experiment. First, it gives me real attendance data without spending class time on a roll call of the whole roster. It also puts some pressure on me to make sure class is worth attending. I will be able to see if we regularly have very low attendance, which is a signal to me that I should probably change something about our class meetings.

This is an experiment and is only 1% of your grade. I will be curious to hear your thoughts and ideas throughout the semester.

Attendance-related accommodations

If you have a Letter of Accommodation from the Disability Resource Center that includes accommodations around attendance, those accommodations take precedence over the attendance and late work policies described here. Please share your LOA with me and we will work out how the accommodations apply to this course.

Student-athletes

UIC has an official missed class time policy under which student-athletes provide formal documentation for class time missed due to NCAA or institutionally sanctioned events. Student Athlete Academic Services (SAAS) will provide you with that documentation; please bring it to me as early as you can so we can plan around it.

Privacy Note (Recording in Class)

As of right now I do not plan to record class sessions this semester. If this changes, I will ask for your permission to record, and it will most likely be only audio.

Similarly, please do not record any video or audio from class without permission.

Please do not wear glasses or other devices that record audio and/or video. If you need to make an arrangement to use a wearable computer of some kind, please come see me.

Classroom Conduct and in-class Technology

TLDR: no phones, no laptops in class - take notes on paper! :-D

This class meets twice a week for one hour and fifteen minutes, totaling 2.5 hours per week. The purpose of our meeting time is to learn, and the following policies are written with that purpose in mind. Outside of our meeting times, of course, you are free to use your devices however you choose.

I remember the frustration and shock I felt in my first Computer Science course in college, when I learned that students were not allowed to use computers in class. I have slowly realized that this policy was best for me as a student, and in most cases, it is the best policy for an effective learning environment.

In Summer 2026, I looked at peer-reviewed scientific evidence on the subject. As with other times I have explored the evidence, my impression was that **in-class device usage is detrimental for learning**, even when students feel accustomed to multitasking. Individual studies as well as meta-reviews continue to characterize devices in class (mainly phones and laptops) as fundamentally distracting, with a negative influence on various measures of learning and performance.

Interestingly, while some students can multitask and still comprehend material *during* class time, those students experience significantly reduced long-term retention (Glass & Kang, 2019).

For these reasons and others, **by default, all class periods will be free from cell phones, laptops, tablets, and wearable computers (i.e. "smart glasses")**. In some cases, we will use devices for activities, usually at the end of class.

The no-phone, no-laptop policy is intended for your own benefit as well as the benefit of your classmates, as using a phone/laptop often has a chain reaction of distracting other students.

For most class sessions, you will be provided a worksheet to take guided notes and write down questions that arise or factoids you wish to research after class. You are always encouraged to ask about factoids that you might otherwise look up on your device - someone in the class might know the answer, or at least have some insight!

Also, if you prefer to take notes on a device that can sit flat on the table, and you are confident that you can do this without distracting yourself and without distracting others, please come see me to make an arrangement.

Finally, **I will revisit the no-phone no-laptop policy with students throughout the semester**, and I remain open to revising the policy based on feedback / student needs.

In case you are curious, here is some of the evidence I looked at during Summer 2026 regarding in-class device distractions:

- Flanigan, Brady, Dai, & Ray (2023). Managing student digital distraction in the college classroom: A self-determination theory perspective. *Educational Psychology Review*. doi.org/10.1007/s10648-023-09780-y
  - "…research has suggested that these digital distractions can negatively impact learning and performance."
- Dontre (2021). The influence of technology on academic distraction: A review. *Human Behavior and Emerging Technologies*. doi.org/10.1002/hbe2.229
  - "The detrimental effects of student smartphone and social media use on academic distraction are more conspicuous, especially with the pervasiveness of personal digital devices."
- Martin, Long, Haywood, & Xie (2025). Digital distractions in education: A systematic review of research on causes, consequences and prevention strategies. *Educational Technology Research and Development*. doi.org/10.1007/s11423-025-10550-6
  - "Consequences for digital distraction included personal performance issues (66.67%), ineffective classroom instruction (23.33%), and problematic technology use (10%)."
- Wang, Tigelaar, Zhou, & Admiraal (2023). The effects of mobile technology usage on cognitive, affective, and behavioural learning outcomes in primary and secondary education: A systematic review with meta-analysis. *Journal of Computer Assisted Learning*. doi.org/10.1111/jcal.12759
  - Notably, this review found that *structured, instructor-directed* technology use can help: "using mobile technology produced medium positive and statistically significant effects on primary and secondary students' learning."
- Sana, Weston, & Cepeda (2013). Laptop multitasking hinders classroom learning for both users and nearby peers. *Computers & Education*. doi.org/10.1016/j.compedu.2012.10.003
  - "…multitasking on a laptop poses a significant distraction to both users and fellow students and can be detrimental to comprehension of lecture content."

A related (and somewhat separate) question is whether writing notes by hand is better than typing them, *even without* any distraction. The evidence is somewhat mixed:

- Craik & Lockhart (1972). Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior*. doi.org/10.1016/S0022-5371(72)80001-X
  - The basic theory: what makes a memory durable is how deeply you processed the material
- Mueller & Oppenheimer (2014). The pen is mightier than the keyboard: Advantages of longhand over laptop note taking. *Psychological Science*. doi.org/10.1177/0956797614524581
  - "…laptop note takers' tendency to transcribe lectures verbatim rather than processing information and reframing it in their own words is detrimental to learning."
- Morehead, Dunlosky, & Rawson (2019). How much mightier is the pen than the keyboard for note-taking? A replication and extension of Mueller and Oppenheimer (2014). *Educational Psychology Review*. doi.org/10.1007/s10648-019-09468-2

- A replication that did not fully reproduce the benefits of handwriting: "concluding which method is superior for improving the functions of note-taking seems premature."

This is one reason I am open to you taking notes on a tablet-like device, if you really want to :-)

Academic Integrity and Assurance of Learning

UIC is an academic community committed to providing an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity. In this community, all members including faculty, administrators, staff, and students alike share the responsibility to uphold the highest standards of academic honesty and quality of academic work so that such a collegial and productive environment exists.

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all of your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code. Unacceptable behavior includes cheating, unauthorized collaboration, fabrication or falsification, plagiarism, multiple submissions without instructor permission, using unauthorized study aids, coercion regarding grading or evaluation of coursework, and facilitating academic misconduct. Please review the UIC Student Disciplinary Policy for additional information about the process by which instances of academic misconduct are handled towards the goal of developing responsible student behavior.

By submitting your assignments for grading you acknowledge these terms, you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone in any way during an exam or other online assessment.

In line with UIC's mission to provide "an environment in which research, learning, and scholarship can flourish and in which all endeavors are guided by academic and professional integrity," **the work you turn in for this class must be your own work**.

This is to ensure that **you** develop skills and knowledge from this course that will be valuable to you in your future courses, your career, and other areas of your life. In other words, the point is to **assure that you are learning** so that UIC fulfills its educational mission.

Academic integrity also includes respecting the copyright of course materials — see "Respect for Copyright" above.

This does not mean you need to complete all work in a bunker. You are encouraged to discuss your ideas with friends and classmates, and to explore related work. An important part of scholarship is standing on the shoulders of giants (and acknowledging whose shoulders we stand upon) — in other words, **cite your sources**.

Please ask the instructor if you need clarity about academic integrity for specific exercises. When in doubt, it is always better to check. I will reach out to you personally before submitting a report of academic misconduct.

Adapted from the UIC CATE syllabus template.

Use of Generative Artificial Intelligence

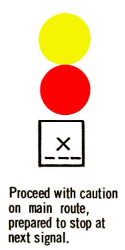
TLDR: wait until later in the semester.

While this class will eventually invite you to experiment with some LLM-based coding tools, we will be working up to that point. The class moves through three phases, named after CTA rail signals:

Double red — “stop and stay”

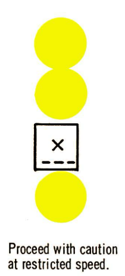
No AI/LLM use at all. Some concepts should be fully learned and absorbed before you reach for a tool. The first few weeks of the semester are double red, with the goal of developing a foundational understanding and discernment.

Yellow over red — “proceed with caution, prepared to stop”



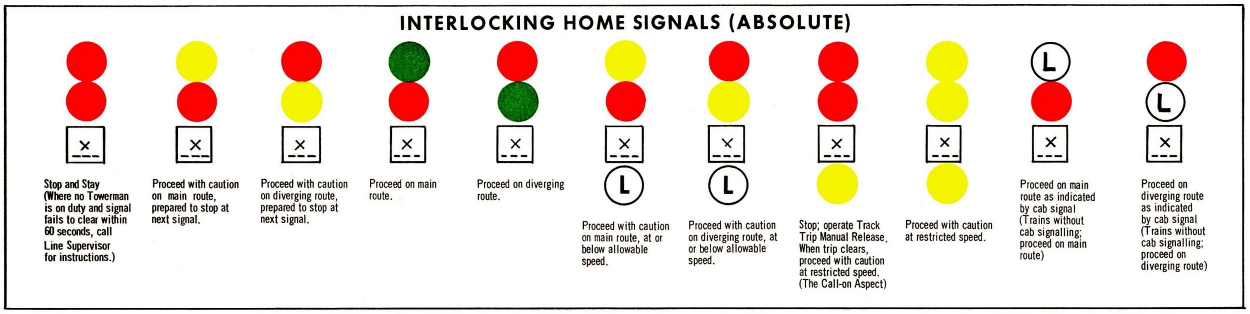
Limited, optional, specific use; use cautiously, with citation. A specific, carefully-scoped set of tasks where AI/LLM use is allowed, but not required, for parts of an exercise. See how the tool fails, decide if and when you find it useful, and pay close attention.

Double yellow — “proceed with caution at restricted speed”



You choose how to use it; use cautiously, with citation. You choose where AI/LLM tools fit into your own workflow. Every use is still at reduced speed and full attention, and you should be prepared to justify it.

Just for fun, here is the full set of eleven signals from the 1973 CTA rulebook:



Eleven signals from the 1973 CTA rulebook.

Each exercise will indicate which of these three applies. Note that use of AI/LLMs is never required. Please ask if you have any questions or uncertainties - again, when in doubt, it is better to check first.

Citing your use of AI/LLMs. Whenever the signal is yellow-over-red or double yellow, if you do use an AI/LLM-based tool, include a citation, and be specific:

> I used [tool and version] to [what you asked it to do]. My prompt was: [prompt, or a summary if it was long]. Here is what I did with the output: [what you kept, changed, or threw away].

This policy is my current best judgment as of August 2026, and it will keep evolving, maybe even during this semester. I am genuinely curious about evidence-based arguments for various different policies - if you see them, bring them to class so we can discuss!

More details on the three phases, including the signal diagrams they are named after, are in the course FAQ. See also the CATE Teaching Guide on Creating AI Policies for Teaching and Learning.

FERPA

Federal law (the Family Educational Rights and Privacy Act) prohibits me from disclosing or discussing your grades over e-mail. Please come see me during drop-in hours or make an appointment if you want to talk about your work. I will work with TAs to try to keep grades updated in Canvas, so you can see how you are performing in class.

Adapted from the UIC CATE syllabus template.

IV. Course Schedule

Weekly Schedule

Week	Class Day	Topic	Before Class	In Class
1	Monday	Course introduction; Data science lifecycle		
1	Wednesday	Asking questions	Homework 1 Due	Week 1 Quiz

Week	Class Day	Topic	Before Class	In Class
2	Monday	Statistics review; Python foundations; Dataframes and Polars	Homework 2 Due	
2	Wednesday	Obtaining data	Project checkpoint: question memo	
3	Monday	n/a	Homework 3 Due	
3	Wednesday	Data formats; Wrangling	(Project groups assembled)	Week 3 Quiz
4	Monday	Exploratory analysis	Homework 4 Due	
4	Wednesday	Descriptive statistics	Coding Exercise 1 Due	
5	Monday	Descriptive visualization	Homework 5 Due	
5	Wednesday	Descriptive visualization	Project checkpoint: data acquisition	Week 5 Quiz
6	Monday	Visualization rhetoric	Homework 6 Due	
6	Wednesday	Visualization rhetoric	Coding Exercise 2 Due	
7	Monday	Hypothesis testing	Homework 7 Due	
7	Wednesday	Estimation; Sampling; Randomness	Project checkpoint: EDA	Week 7 Quiz
8	Monday	Linear regression	Homework 8 Due	
8	Wednesday	Classification	Coding Exercise 3 Due	
9	Monday	Decision trees	Homework 9 Due	
9	Wednesday	Support vector machines; Kernels	Project checkpoint: data documentation	Week 9 Quiz
10	Monday	Clustering; Principal component analysis	Homework 10 Due	
10	Wednesday	Model evaluation	Coding Exercise 4 Due	
11	Monday	Recommendation systems	Homework 11 Due	
11	Wednesday	A/B testing	Project checkpoint: rhetorical analysis	Week 11 Quiz
12	Monday	Network analysis	Homework 12 Due	
12	Wednesday	Graphs; Network measures	Coding Exercise 5 Due	
13	Monday	Topic TBD (flex day)	Homework 13 Due	
13	Wednesday	Topic TBD (flex day)	Project checkpoint: written report	Week 13 Quiz
14	Monday	Project presentations		
14	Wednesday	n/a		
15	Monday	Project presentations		

15	Wednesday	Project presentations

Academic Deadlines

See the UIC Academic Calendar for add/drop, withdrawal, and other academic deadlines.

V. Accommodations

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, you should connect with the Disability Resource Center (DRC) at drc.uic.edu, drc@uic.edu, or at (312) 413-2183 to create a plan for reasonable accommodations. In order to receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Letters of Accommodations can be presented at any point of the semester and, unless otherwise noted, do not expire once obtained. Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

I will make every effort to avoid scheduling exams or requiring student projects be submitted on religious holidays. If you wish to observe your religious holidays, please notify me by the tenth day of the semester of the date when you will be absent unless the religious holiday is observed on or before the tenth day of the semester. In such cases, please notify me or a TA as soon as possible to let me know when you will be absent.

I will make every reasonable effort to honor your request and not penalize you for missing the class. If an examination or project is due during your absence, you will be given an exam or assignment equivalent to the one completed by those students in attendance. Students may appeal through campus grievance procedures for religious accommodations.

Much of the remaining syllabus was copied or lightly modified from the UIC CATE syllabus template.

VI. Classroom Environment

Inclusive Community

At UIC we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If there are aspects of the instruction or design of this course that result in barriers to your inclusion, engagement, accurate assessment or achievement, please notify me as soon as possible.

Preferred Names and Pronouns

If your name or pronouns do not match the class roster, please let the class know so that we can address you the way you want to be addressed. I use he/him pronouns. For more information, see: mypronouns.org/what-and-why.

Community Agreement

This will be discussed and developed collectively in class.

VII. Feedback & Support

Providing Feedback About This Course

This is my first time teaching a full-semester data science course, and I am confident that there will be many ways to improve the course experience. I will provide structured opportunities for you to share feedback, which I take seriously – each semester I make changes based on feedback from previous students. I also try to make time in each class session for students to ask any questions they may have about the course.

You are also encouraged to edit the course site by submitting fixes/improvements directly in GitHub, or opening an issue on the GitHub page. For details about contributing, see the CONTRIBUTING.md file.

UIC Resources

As a student at UIC, you may experience challenges related to academics, finances, student life, and/or your personal well-being. Please know this is completely normal and that you shouldn't hesitate to ask for help. You are welcome to come to me, your college advisors, or any number of other support services and resources available to all UIC students:

- Current Student Resources — a one-stop shop for links to resources in the following categories: General, Academic, Student Support, Student Life, Technology, Health and Safety, and Getting Around Campus
- UIC Tutoring Resources
- UIC College of Engineering Tutoring Program
- The UIC Library provides access to resources, study rooms, and research support both online via chat and in person; see also the subject-specific Research Guides
- Programs and initiatives supporting the UIC undergraduate experience and academic programs through the Student Success Initiative
- If you are experiencing personal hardships and need resources and assistance, please reach out to the Student Assistance division, supported by UIC's Office of the Dean of Students.
- Student Guide for Information Technology — a comprehensive resource for UIC students describing the most commonly used IT services and tools supporting your success
- Navigating Health Resources at UIC — explore the many resources on campus to support students' health and wellbeing and learn about how you can access them

Importantly, if you are in immediate distress, please know the UIC Counseling Center has crisis intervention services available 24 hours a day, 7 days a week. You can walk into the Counseling Center during business hours to request to talk to a crisis counselor, or you can call the Counseling Center main line at (312) 996-3490. If you are calling outside of business hours, select "2" during the automated greeting to be connected to a crisis counselor. You can find additional mental health resources and services on the Counseling Center website and in the Student's Guide to Accessing Behavioral Health Services at UIC. For immediate crisis support outside of UIC, call or text "9-8-8" any time, 24 hours a day, 7 days a week. Dialing 988 will connect you to a trained crisis professional, and the 988 website has an online chat feature too.

The Writing Center offers friendly and supportive tutors who can help you with reading and writing in any of your courses, not just English. For more information and to schedule an appointment, visit the Writing Center website.

The Math and Science Learning Center (MSLC), located in the Science and Engineering South Building (SES) at 845 W. Taylor St. 3rd Floor, Room 247, offers tutoring and study space for students in Math, Biological Sciences, Chemistry, Earth and Environmental Sciences, and Physics. Visit the MSLC website, call 312-355-4900, or email mslc@uic.edu.

The Academic Center for Excellence (ACE) can help if you feel you need more individualized instruction in reading and/or writing, study skills, time management, etc. Please call (312) 413-0031 or visit the Academic Center for Excellence (ACE) website for more information.

Supporting Parenting Students: UIC supports parenting students by providing affordable, high-quality childcare, a wraparound care system, study-play areas, accessible lactation rooms, and family events through the Little Sparks Program. For more information, contact Melinda Young (mmcghe2@uic.edu) or visit the Little Sparks website.

Student Health Services are available if you get sick during the semester. The Family Medicine Center has attending physicians, residents, and nurse practitioners on staff to provide the full range of primary health care services, and offers a limited set of services at no additional cost to students who pay the Student Health Service Fee (SHSF), including care for an acute or urgent illness (e.g. cold, strep throat, asthma attack, flu) or an injury. More information is on the Family Medicine Center's Student Health webpage.

Campus Safety: The UIC Safe App (link to download) provides on-campus emergency contact with dispatchers and first responders. If you are uncomfortable traveling anywhere on campus, use the Walking Safety Escort and/or Night Ride. Emergency response guides can be found at the Office of Preparedness and Response. In an on-campus emergency, dial 5-5555 from a campus phone, or (312) 355-5555 from a cell phone.

Title IX / Campus Advocacy Network: For confidential victim services and advocacy, contact UIC's Campus Advocacy Network at 312-413-8206. To make a report to UIC's Title IX office, email TitleIX@uic.edu or call (312) 996-8670.

VIII. Disclaimer

This syllabus is intended to give guidance on what may be covered during the semester and will be followed as closely as possible. However, the instructor reserves the right to modify, supplement, and make changes as course needs arise. Significant changes will be communicated in advance as in-class announcements and in writing via Canvas Announcements and email.

Parts of this syllabus were copied or lightly modified from the UIC CATE syllabus template.